

LESSON 6.2

CURRENCY CONVERSIONS



LESSON INTRODUCTION

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

LEARNING TARGETS

- I can convert between two different forms of currency.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- What is a unit rate? What is an exchange rate? How are they alike or different?
- How do you use a unit rate to convert between two currencies?
- How do you determine which currencies are more valuable than others?

LESSON NARRATIVE

Students continue their journey from Lesson 1 through the land of Mathlandia and its currencies. While students engaged primarily with conversions in Lesson 1, this lesson begins incorporating problem-solving situations in which students begin comparing and contrasting different currencies using mathematical reasoning and thinking. Both “unit rate” and “exchange rate” are primary vocabulary of communication throughout this lesson.

COMPONENT SUMMARY

6.2.1 WARM-UP (~5 MIN.)

Compare and contrast two currencies using a table with given unit rates

6.2.3 GUIDED INSTRUCTION (10 MIN.)

Use a currency chart to convert currencies

6.2.5 YOUR TURN! (10 MIN.)

More practice in currency and exchange rates in computation and one problem-solving situation

6.2.2 COLLABORATION (10 MIN.)

Explore the use of unit rate when converting between currencies

6.2.4 GUIDED PRACTICE (10 MIN.)

Practice in currency and exchange rates in problem-solving situations

6.2.6 WRAP-UP (~5 MIN.)

Application problems with dollars/francs (real-world) and op/peaks (Mathlandia)

RESOURCES

GLOSSARY

Key Terms:

- Exchange rate

Additional Terms:

- Unit rate

MATERIALS

- (Optional) Mathlandia currency
- (Optional) Foldable or graphic organizer

ADDITIONAL PREPARATION

- Print Mathlandia currency to use in class (if applicable).
- Consider a chart or graphic organizer to display the exchange rates in 6.2.3 Guided Instruction.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle interpreting which currency would be the numerator and which would be the denominator, often taking the reciprocal instead.
- Student computational errors might detract from conceptual understanding.
- Students may struggle with operations with decimals.

THINGS TO CONSIDER

- Prompt students to write the units in their exchange rate. Labeling units with values from beginning to end will encourage more precision in the process and solution.
- Ask students to consider if their answers make sense for every activity. Should they have more or less of the other currency?
- Consider providing calculators for students to use when computing conversions.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Students practiced converting between currencies in Lesson 1, but the 6.2.1 Warm-Up in this lesson will be the first time they formally compare and contrast using an exchange rate. Think-Pair-Share provides an opportunity for students to make sense of the information through three phases that encourage sense-making, confidence, and collective knowledge sharing.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

This lesson presents a multitude of opportunities for rich dialogue on real-world applications of math and project-based learning. The majority of the lesson continues with applications of fake currency in Mathlandia; however, in planning, consider what a project with real-world currency might look like for your students, using the same guiding prompts from the lesson.

DIFFERENTIATE INSTRUCTION

If you have students who have lived in or visited countries with different currencies, consider how incorporating their stories and perspectives into this lesson might engage a wider lens of learners. This lesson provides both actual and fake currencies, so consider the lived experiences of your students when planning which content you may incorporate and why.

Use printable options for the fake currency incorporated into this lesson. Consider how currency manipulatives might enhance this lesson for kinesthetic and visual learners.

Consider an alternative performance task in which students create their own market shop of items and must provide a menu listed in at least three different currencies. The agency of the task will engage a variety of learners while still challenging students to apply concepts of unit rates and conversions.

6.2.1 WARM-UP: WOULD YOU RATHER?

TEACHER MOVES

The previous lesson explored exchange rates with students applying ratios to determine currency value. This will likely be the first time students are being asked to interpret which is more valuable through the lens of this representation.

TEACHER MOVES

Think-Pair-Share

Consider conducting the Think-Pair-Share instructional routine for this Warm-Up. Students will make better sense of thinking and reasoning with the opportunity to independently process and then refine answers with a partner before the teacher elicits more precise responses as a whole group.

DIFFERENTIATE INSTRUCTION

This activity is more of an exercise in interpreting data by means of this table than comparing exchange rates. Provide scaffolding questions or sentence stems for interpreting the table for students to effectively engage with. • For every 1 sal, I can get ____ dollars.

- For every 1 dollar, I can get ____ sal.
- What can you buy more with, 1 sal or 1 dollar?
- Which is more valuable?
- Which would you rather have? Why?



6.2.1 Warm-Up: Would you Rather?

Consider the exchange rates from currency in a Mathlandia country, Scalenia, compared to the United States (U.S.) dollar.

	U.S. dollar (\$) Equivalent	Currency per U.S. dollar (\$)
Scalenia 1 sal (\$)	1.5939	0.6274

1. Would you rather have \$100 or \$100? Explain your reasoning.

Answers will vary.

Example response: 1 sal (\$) is worth \$0.59 more per U.S. dollar.

$$\$100 \times \frac{\$1.5939}{\$1} = \$159.39$$

$$\$100 > \$100$$



6.2.2 Collaboration: Reasoning and Sense Making

Work with your partner on the following problems.

- The exchange rate used to change Multiplicity macs (m) to Proportia pops (p) is $\frac{5}{8}$. Complete the statement.

For every 8 mac(s) there are 5 pop(s).

- On October 28, 2020, Lorenzo's family exchanged 250 U.S. dollars for 5,330 Mexican pesos.

U.S. dollars	Mexican pesos
250	5,330
25	533
2.50	53.3

- Complete the table.
- Complete the rate to represent the exchange rate of Mexican pesos per U.S. dollar on October 28, 2020.

$$\frac{21.32 \text{ pesos}}{1 \text{ US dollar}}$$
- Lorenzo bought a blanket for 247 pesos that day. Without calculating, estimate the cost of the blanket in dollars. *Answers will vary.*
 $\approx \$10$
- What was the cost of the blanket in dollars?

$$247 \text{ pesos} \times \frac{\$1}{21.32 \text{ pesos}} = \$11.59$$
- What was the exchange rate in terms of U.S. dollars per peso on October 28, 2020? Write your answer as a decimal to the nearest hundredth.

\$0.05 U.S. dollars per Mexican peso

$$\frac{\$250}{5330 \text{ pesos}} \div \frac{5330}{5330} = \$0.05$$

6.2.2 COLLABORATION: REASONING AND SENSE MAKING

QUESTIONING

Looking for a way to bring this lesson to life? Visit our Teacher Area for printable versions of Multiplicity macs and Proportia pops to engage students with manipulatives that can bring this currency and activity to life.

ENRICHMENT

- What is worth more: 250 U.S. dollars or 5,330 Mexican pesos?
- What do you notice about the relationship between the dollar and the peso?

TEACHER MOVES

Students will formalize the process of converting currencies in the next component. This component is meant to give students an opportunity to collectively solve problems using their prior knowledge of unit rates.

DIFFERENTIATE INSTRUCTION

Do you have students who have used pesos or lived in countries with different currencies? This portion of the lesson is an opportunity to incorporate their stories into the lesson and differentiate some of the content of the lesson based on students' lived experiences.

6.2.3 GUIDED INSTRUCTION: CONVERTING CURRENCIES

TEACHER MOVES

“Exchange rate” here will be students’ formal introduction to this term. Consider giving students the opportunity to make sense of the phrase through one of the previous activities with questions like “What was the exchange rate between sals and dollars? What was the exchange rate between pesos and dollars?”

QUESTIONING

Add complexity to this activity by having students write an inequality statement using all forms of currency.

$\text{var} > \text{pop} > \text{equi} > \text{sals} > \text{ops} > \text{peaks}$

What is the least valuable currency? Most valuable?

6.2.3 Guided Instruction: Converting Currencies



An **exchange rate** is a ratio that compares two different currencies. An exchange rate can be written as a fraction, ratio, or decimal.

When the exchange rate is written as a unit rate, it shows the value of a certain currency for each one unit of the other currency.

Ratio tables, proportions, or other equations are useful when converting one currency to another.

Recall the exchange rates from the Mathlandia Exploration.

	1.32	sals, \$
Equivaland	0.85	pop, ₱
1 equi, €	15.90	peaks, ₤
	0.30	var, ⱪ
	7.75	ops, ₤

- Determine the values for each exchange.
 - 45 equis = 38.25 pops $45 \text{ equis} \times \frac{0.85 \text{ pops}}{1 \text{ equi}}$
 - 1,113 peaks = 70 equis $1,113 \text{ peaks} \times \frac{1 \text{ equi}}{15.9 \text{ peaks}}$
- Emma travels to Scalenia from Equivaland. She exchanges 200 equis. How much does she receive in sals?

$$200 \text{ equis} \times \frac{1.32 \text{ sals}}{1 \text{ equi}} = \boxed{264 \text{ sals}}$$
- Emma purchases these items while in Scalenia. What is the cost of each item in equis, €?
 - A necklace for \$64.35

$$64.35 \text{ sals} \times \frac{1 \text{ equi}}{1.32 \text{ sals}} = 48.75 \text{ equis}$$
 - A scarf for \$11.22

$$11.22 \text{ sals} \times \frac{1 \text{ equi}}{1.32 \text{ sals}} = 8.5 \text{ equis}$$

6.2.4 GUIDED PRACTICE: CONVERTING CURRENCIES

ENRICHMENT

- How could unit rate be used in this situation?
- Is this question asking for the number of euros or the number of dollars?
- How should the unit rate representing the exchange rate be set up? Which currency should be in the numerator and denominator?
- How do you know it will be more?
- What are more valuable, euros or dollars? If euros are more valuable, then wouldn't the shirt be less valuable in dollars, therefore cost less in dollars?

6.2.4 Guided Practice: Converting Currencies

- Jaxon exchanged 500 dollars for 420 euros.
 - What was the exchange rate in euros per dollar?

$$\frac{420 \text{ euros}}{500} \div \frac{500}{500} = \frac{0.84 \text{ euros}}{\$1}$$
 - Jaxon paid 16.49 euros for a shirt. Do you expect the price of the shirt to be more or less than 16.49 dollars? Explain your thinking without doing any calculations.

Answers will vary.
It will be more than \$16.49 because 1 euro is worth more than \$1.
 - What is the price of the shirt in dollars?

$$16.49 \text{ euros} \times \frac{\$1}{0.84 \text{ euros}} = \boxed{\$19.63}$$
- Keisha travels from Proportia to Variableya. The currency exchange rate is 1 pop = 0.35 var. Keisha exchanges 120 pops. How much does she receive in vars?

$$120 \text{ pops} \times \frac{0.35 \text{ var}}{1 \text{ pop}} = \boxed{42 \text{ vars}}$$

**6.2.5 Your Turn!**

Recall the exchange rates from the Mathlandia Exploration.

Scalenia 1 sal, §	≈	0.64	pop, ₱
		12.05	peaks, Å
		0.76	equi, €
		0.23	var, ⋈
		5.88	ops, ⊕

- Determine the equivalent values for each currency exchange.

$$61 \text{ sals} = \underline{14.03} \text{ vars} \quad \underline{140} \text{ sals} = 106.40 \text{ equis}$$

$$18 \text{ peaks} = \underline{1.49} \text{ sals} \quad 90 \text{ sals} = \underline{529.2} \text{ ops}$$

$$61 \text{ sals} \times \frac{0.23 \text{ var}}{1 \text{ sals}} = 14.03 \text{ var} \quad 106.40 \text{ equis} \times \frac{1 \text{ sal}}{0.76 \text{ equis}} = 140 \text{ sals}$$

$$18 \text{ peaks} \times \frac{1 \text{ sal}}{12.05 \text{ peaks}} = 1.49 \text{ sals} \quad 90 \text{ sals} \times \frac{5.88 \text{ ops}}{1 \text{ sals}} = 529.2 \text{ ops}$$

- Logan lives in Equivaland and wants to take his family on vacation.

Logan has saved up 400 equis for the trip. Aunt Emma, from Prismopolis, sent 1,200 peaks to help pay for the trip. The exchange rate is 1 equi = 15.9 peaks.

With Aunt Emma's gift, how much does Logan have available for the trip in equis? Round to the nearest hundredth.

$$1200 \text{ peaks} \times \frac{1 \text{ equi}}{15.9 \text{ peaks}} = 75.47 \text{ equis}$$

$$75.47 \text{ equis} + 400 \text{ equis} = 475.47 \text{ equis}$$

**6.2.6 Wrap-Up: Ticket Out the Door**

- Milo exchanged \$400 for 367 francs. Complete the two statements based on this currency exchange. Rounded answers are accepted.

$$1 \text{ dollar} = \underline{0.9175} \text{ franc(s)} \quad \frac{367 \text{ francs}}{\$400} \div \frac{\$400}{\$400} = \frac{0.9175 \text{ francs}}{\$1}$$

$$1 \text{ franc} = \underline{\$1.09} \text{ dollar(s)} \quad \frac{\$400}{367 \text{ francs}} \div \frac{\$367}{\$367} = \frac{1.09}{1}$$

- The exchange rate from Integerina ops to Prismopolis peaks is 1 op = 2.05 peaks.

Ava exchanged 500 peaks for ops. How many ops did she receive?

$$500 \text{ peaks} \times \frac{1 \text{ op}}{2.05 \text{ peaks}} = 243.90 \text{ ops}$$

6.2.5 YOUR TURN!**DIFFERENTIATE INSTRUCTION**

Consider an alternative performance task that combines the Your Turn! and 6.2.6 Wrap-Up for a differentiated approach to students producing applications in this lesson.

For examples on how to differentiate this activity, see the differentiation note for the 6.2.6 Wrap-Up!

6.2.6 WRAP-UP: TICKET OUT THE DOOR**DIFFERENTIATE INSTRUCTION**

Consider an alternative performance task that combines the 6.2.5 Your Turn! and Wrap-Up for a differentiated approach to students producing applications in this lesson.

Examples:

- Students need to tell a story based on the four conversions presented in question 1.
- Students create their own market shops based on the conversions presented or come up with their own values and present menus of items that include at least two conversions.
- Students take the chart from the 6.2.1 Warm-Up to convert peaks, sals, or equi into at least two other real-world currencies.